MIT ZARAGOZA
International Logistics Program

MZ505 Inventory Systems 2026/2027

Assignment #1

Prof. Spyros Lekkakos

- **Individual Assignment**
- **Due: Sep 21st (Monday): 08.30 am**
- **Please submit you work online via Canvas (only one .pdf file) and provide a hard copy before class.**

General Remarks

- Reasonably sufficient justification of your work is required to get the full credit.
- If you submit hand-written answers, please make sure that the document is well-structured and clear (points may be deducted for poor clarity).

Note: Assume (if needed) 1 year = 12 months = 52 weeks = 365 days

Problem 1. Finite production (Max Points 30)

Hyundai's SUV division is selling only one product and is running an independent, fully automated facility with a production rate of 150 cars per month. The company needs 3 days of setup period to warm up the facility, and €1,000 is incurred daily for this task. It is facing a stable customer demand of 1,200 cars per year. The unit production cost is €15,000 and the selling price is €25,000. The company is applying an annual unit holding cost of 15% of product cost.

(a) How many cars shall Hyundai produce whenever it starts the facility to maximize the total profit in the long run? (5 points)

(b) How many cars shall the company produce whenever it starts the facility to minimize the total cost in the long run? (5 points)

(c) How many days per year shall the facility be idle for (b) above? (5 points)

(d) What is the average inventory level for (b) above in the long run? (5 points)

(e) How much is the average total profit per year in the long run? (5 points)

(f) Now the company has a new constraint on the production batch size and the production quantity should be a multiple of 40, i.e., 40-80-120-160-…, whenever it runs the facility. How much is the optimal production quantity to minimize the long-run cost? (5 points)

Problem 2. All unit and marginal unit discounts (Max Points 30)

Ericson's mobile communications division, a smartphone manufacturer, uses 20,000 memory chips per month. Their transportation service provider charges Ericson €300 per shipment, independently of the quantity purchased. The manufacturer offers an all unit quantity discount with a price of €1 per memory chip for orders up to 18,000 units, €0.96 per chip for orders between 18,000 and 36,000 units, and €0.92 per chip for orders larger than 36,000 units. Ericson incurs an annual unit holding cost of 15% of product cost.

(a) What is the optimal lot size for Ericson? (5 points)

(b) What is the annual cost of such a policy? (5 points)

(c) What is the cycle inventory of memory chips at Ericson? (5 points)

[(d)-(e)] The manufacturer now offers a "*marginal*" unit quantity discount for the chips. The first 18,000 chips of any order are sold at €1 per unit, the next 18,000 units are sold at €0.96 and any quantity over 36,000 chips is sold for €0.92 per unit.

(d) What is the optimal lot size for Ericson given this pricing structure, what is the resulting cost, and how does it compare with that in (a) and (b)? (10 points)

(e) What is the cycle inventory of memory chips at Ericson given the ordering policy? (5 points)

Problem 3. Shipment aggregation (Max Points 20)

LG, a refiner in Barcelona, serves three customers near Zaragoza, and maintains consignment inventory (owned by LG) at each location. Demand at the large customer is 48 tons a year, demand at the medium customer is 24 tons per year, and demand at the small customer is 12 tons per year. Product cost for LG is €80,000 per ton and the annual unit holding cost has been estimated to 15% of product cost. Currently, LG uses full-truckload transportation to deliver separately to each customer. Truck capacity is 12 tons and each delivery costs €700 plus €150 per stop (thus, delivering to each customer separately costs €850 per truck). LG is considering aggregating deliveries to Zaragoza on a single truck.

(a) What is the annual transportation and holding cost if LG ships a full truckload each time a customer is running out of stock? How many days of inventory[1] are carried at each customer under this policy? (5 points)

(b) What is the optimal delivery policy to each customer if LG ships separately (not necessarily full truckloads) to each of them? What is the annual transportation and holding cost? How many days of inventory are carried at each customer under this policy? (5 points)

[1] Days of inventory = days of demand that can be met by average (cycle) inventory

(c) What is the optimal delivery policy to each customer if LG aggregates shipments to each of the three customers on every truck that goes to Zaragoza? What is the annual transportation and holding cost? How many days of inventory are carried at each customer under this policy? (10 points)

Problem 4. Power of two policies (Max Points 20)

EasyWash is a washing machine manufacturer operating in Europe. One of its best-selling models is produced in the firm's manufacturing site in Zaragoza and consists of three core subassemblies: rotor, cylinder, and frame. While the rotor is manufactured in-house, the other two subassemblies (cylinder and frame) are sourced from Asia. The cost for placing an order with the cylinder supplier is 2,500€, while the same cost for the frame is 600€. Also, the annual cost of holding one unit of inventory is equal to 75€ for the cylinder and 50€ for the frame. The annual demand for the particular model is equal to 1,200 units (each machine consists of one rotor, one cylinder, and one frame).

Assuming that availability of rotor is not an issue, the firm would like to coordinate the procurement of the other two subassemblies (cylinder and frame) for logistics planning purposes, by using "power of two" policies. What would be the optimal policies for the two subassemblies, considering that the firm's planning team uses one week as the base planning period (T_B)? For each subassembly separately, please evaluate your policy's performance (cost-wise) compared to the optimal one.

QUIZ PROBLEMS

Quiz Problem 1. Batches Production (Max Points 10)

In the Economic Production Quantity (EPQ) model, we have assumed that output immediately becomes available to meet demand. Therefore, we can wait until inventory drops to zero before beginning production. In many cases, however, the finished product cannot be used until an entire batch is complete, perhaps because the batch must be moved to a warehouse, or because some finishing operation (e.g., cooling) is required. Let D = demand rate, P = production rate (P>D), S = setup cost, H = unit holding cost, and Q = production quantity per production run.

(a) Show that the average inventory, optimal production quantity, and total variable cost are respectively equal to:

$$\bar{I} = \frac{1}{2}\left(1+\frac{D}{P}\right)Q, \quad Q^* = \sqrt{\frac{2DS}{H\left(1+\frac{D}{P}\right)}}, \quad \text{and} \quad TVC(Q^*) = \sqrt{2DSH\left(1+\frac{D}{P}\right)}$$

(Hint: consider, to avoid running out of stock, what are the maximum and minimum inventory values for this system). (7 points)

(b) Notice that when output is immediate available, the total variable cost is increasing in the production rate, while in this case (i.e., when the finished product cannot be used until an entire batch is complete), the total variable cost is decreasing in the production rate. Explain the possible reason. (3 points)

Quiz Problem 2. EOQ with Backorders (Max Points 10)

In the Atlantic Coast Tire Corporation (ACT) problem that we discussed during our first lecture on EOQ, we ignored the cost of being out of stock ($7.50 per tire short). Consider the penalty associated with stocking out as cost per unit short per year (i.e., the stockout cost penalizes also the duration of the stockout situation; hence it shall be treated as the holding cost).

Assuming that all demand that is not satisfied on time is backordered (i.e., it is served, usually first, during the next batch delivery):

(a) What is the EOQ for ACT in this case? Please show the steps you followed in deriving this quantity. (7 points)

Hint: Set B the quantity backordered. Make a graph of how the policy will look like if backorders are allowed. Compute the time lengths T_1 (demand is fully met) and T_2 (demand is backordered) as functions of D, Q, and B. Compute average inventory and average backorders and construct the cost function (which is now a function of both Q and B). You can assume that the cost function is jointly convex to Q and B (i.e., you don't need to show second order derivative calculations). Then, take the partial derivatives of the cost function over Q and B. This will result to a system of two equations with two unknowns. By solving this system, you can obtain the optimal Q and B.

(b) Do you expect the new EOQ to be different (smaller or larger) from the classical EOQ derived in class? Please explain why. (3 points)